

MAST32001 Week 4 Summary

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2026-09-25

This week we used the bootstrap for hypothesis tests and regression. Then we moved on to cross-validation.

The main idea for a bootstrap test is simple. We have to resample as if the null hypothesis were true. We can do this in two ways. First, we can fit a model under the null. Then, we simulate new data from that model. Second, we can shift the sample. We make its mean match the null value. Then we resample. We also compared this with permutation tests. Permutation tests work because data have symmetry. They do not need us to estimate anything.

For regression, we had two main options. The residual bootstrap keeps the x value fixed. It only shuffles the errors. This method makes sense when we choose the x values ourselves. It also makes sense when all errors have the same spread. The pairs bootstrap shuffles whole (x, y) pairs together. This method is better when the x values are random. Sometimes the errors do not have the same spread. This problem is called heteroscedasticity. Then the residual bootstrap does not work well. We learned about the wild bootstrap for this case. The wild bootstrap keeps each error tied to its own x . It just flips the error sign. Then we looked at time series. The usual bootstrap breaks time series. It mixes up the order. The block bootstrap fixes this problem. It resamples whole chunks rather than individual points.

The second lecture was about prediction. It asked how well a model predicts data it has not seen. The main warning is that training error lies. If we score a model on the same data we trained it on, it looks better than it actually is. Prediction error splits into three parts. The first part is noise. We cannot get rid of noise. The second part is bias. Bias happens when the model is too simple. Third part is variance. Variance happens when the model jumps around a lot if the data change. The more flexible the model, the lower the bias but the higher the variance, so test error ends up U-shaped.

Then we learned cross-validation. We split the data into folds. We train on most data. We test on the held-out fold. We rotate the folds. Then every point gets tested once. Leave-one-out is the extreme version. And for linear regression, there is a shortcut so we do not have to refit the model n times.

Model selection was the part that took me a whole while. Picking the “best” model with CV adds its own kind of overfitting. If we try 100 models, the winner partly got lucky. There are fixes. One fix is nested cross-validation. Another fix is to make sure test data never leaks into preprocessing. Another fix is to use AIC or BIC. These methods charge extra for each parameter we add. We also saw that the CV setup must match the problem. For time series, we can only predict forward. For data that comes in groups, we hold out whole groups.

Honestly, the easiest thing this week was writing the residual and pairs bootstrap in R. It is a short recipe and the plots are easy to read. The hardest thing was getting my head around why the smallest CV score is not trustworthy on its own.